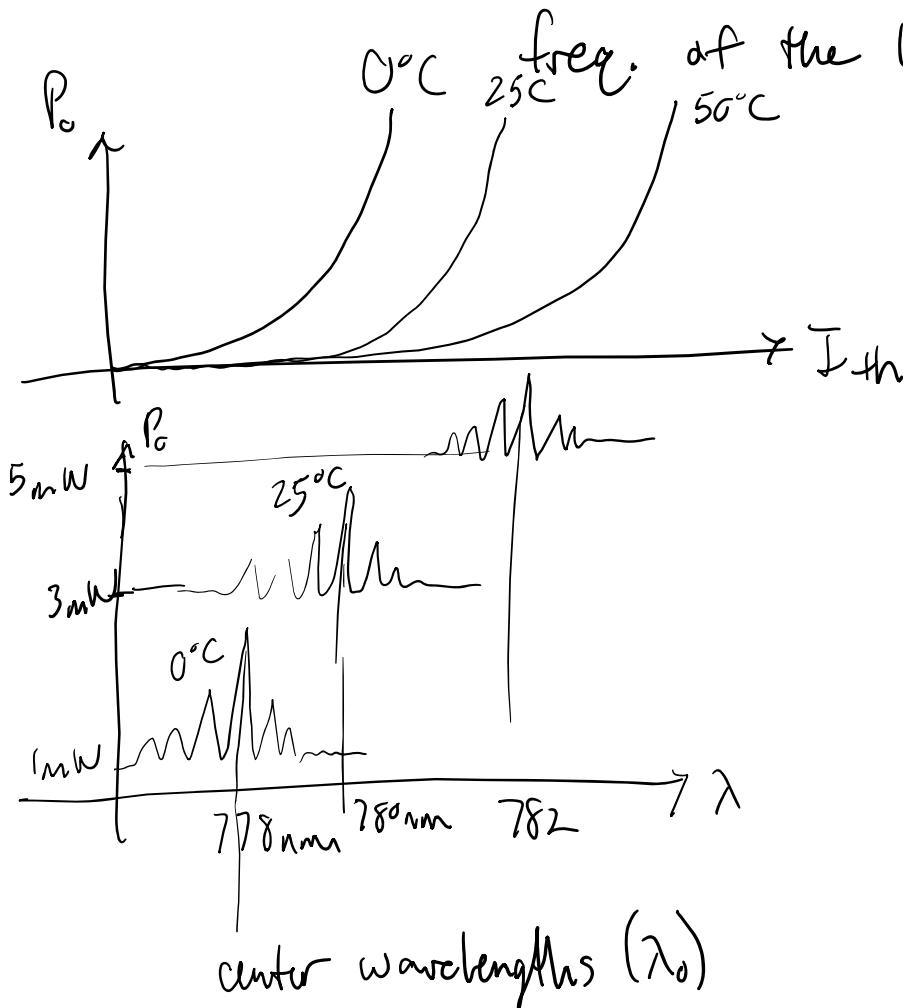


Temperature dependence of Laser diodes

$I_{th} = A e^{(T/T_0)}$; T_0 is the characteristic Temperature or temp that will change the "mode"



* I_{th} is the breakeven Point where beyond the Laser diode will Lose.

* temp affects output λ_0

PN length

LED $\rightarrow$ Modify edges of the chip $\rightarrow$ Laser diode

LED $\rightarrow$ shine light on an LED; it will generate Voltage

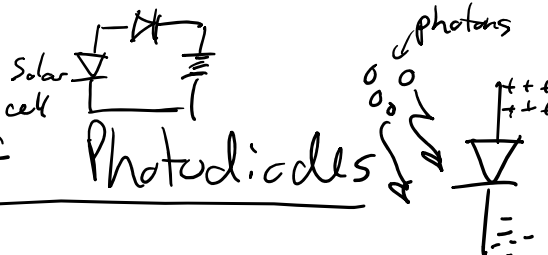
Photodiode $\rightarrow$ Respond fast (w/ lower Voltage)

Solar cells $\rightarrow$ Respond Slow (w/ greater Voltage)

"Fast speed"

• photo voltaic opto-isolators/triggers $\rightarrow$ are diodes

• photo conductors/photo resistors: 
 $\hookrightarrow$ Not diodes

Quantum Efficiency of Photodiodes 

$\eta_e = \frac{\text{\# of collected electrons at the detector's terminals}}{\text{\# of incident Photons}}$

$\eta_e = \frac{(I_{ph}/q)}{(P_D/hf)}$ • P_D is the incident power on the diode (array); taking the diode (array) area into consideration.

• I_{ph} is the current created by the light falling on the diode. (assuming load)

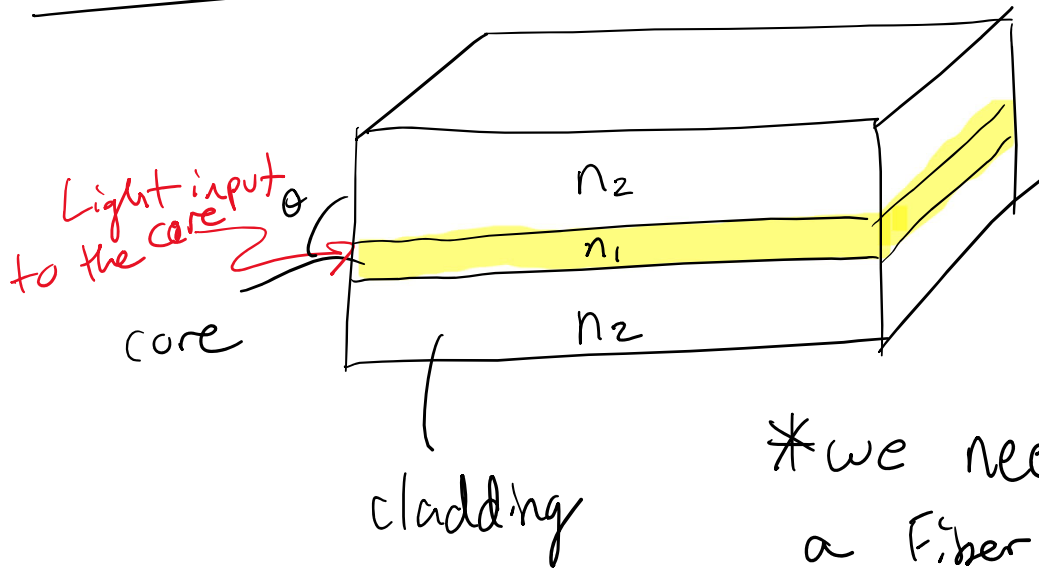
$$\eta_e = \frac{(I_{ph})(hf)}{q P_D}$$

In the sensor domain

$$\text{Responsivity } (R) = \frac{I_{ph}}{P_o}$$

$$R = \eta_e \left(\frac{q \lambda}{hc} \right)$$

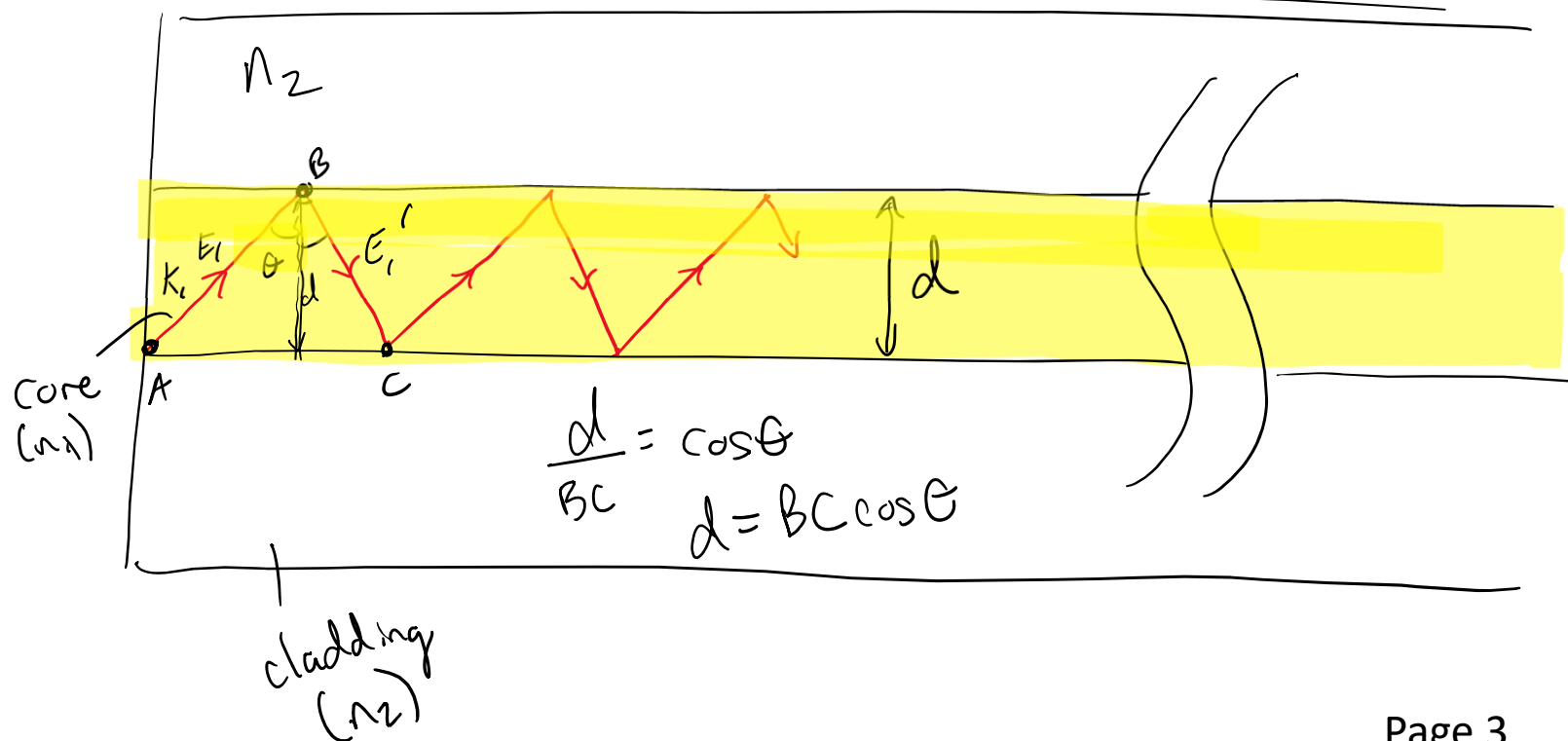
Fiber Optics (optical waveguides)

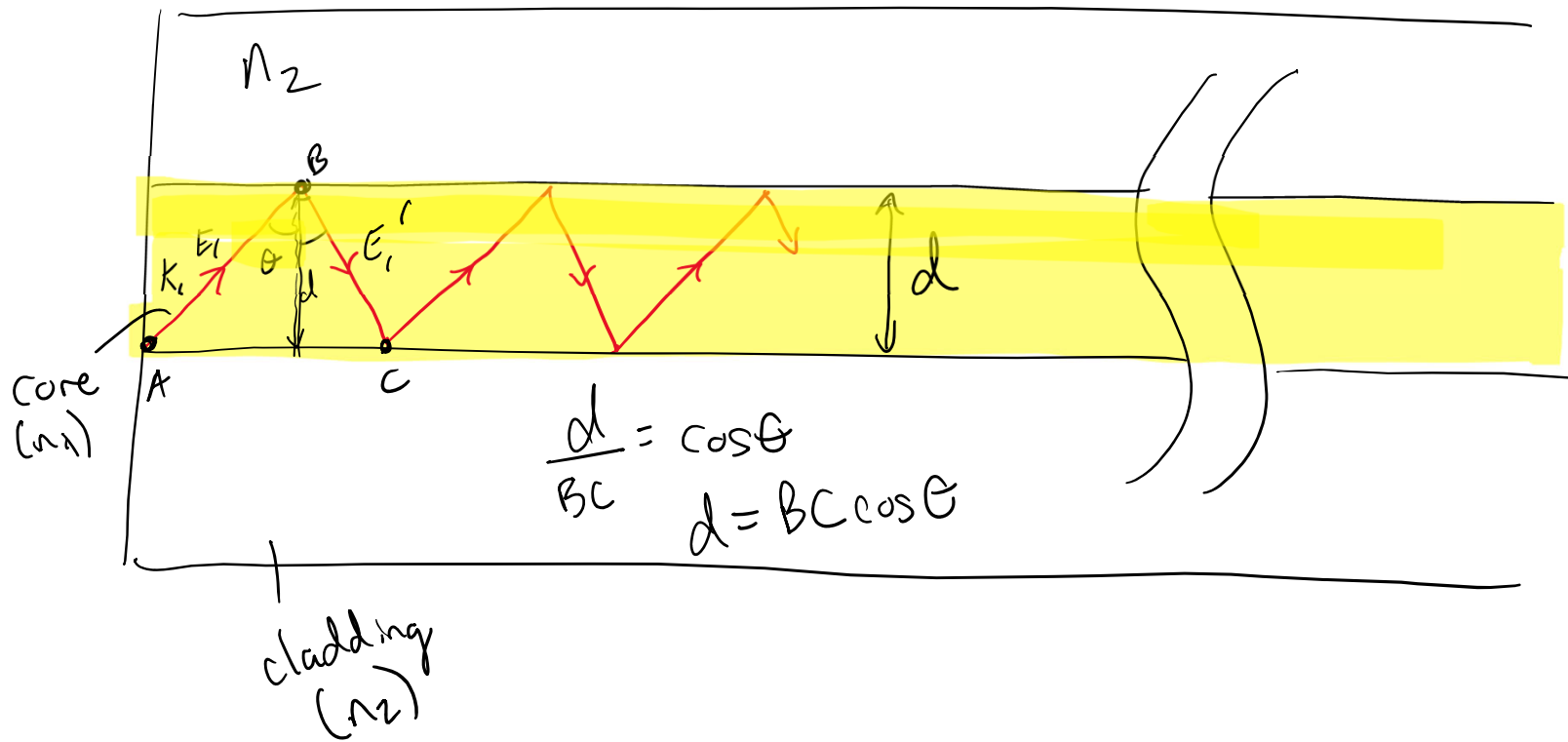


* we need T.I.R. for a Fiber optic to work

$$* \underline{n_1 > n_2}$$

core must have a higher index of refraction than the cladding





the oscillation phase (ϕ) b/w A to B, and B to C
 E_1 will have a slight phase change with E_1'

Related by the path

$$k_1 = k_0(n_1) = \frac{2\pi(n_1)}{\lambda_0}$$

$$\Delta \Phi_{AC} = k_1(AB + BC) - 2\phi = m(2\pi); m = 0, 1, 2, 3, \dots$$

mode (vertical direction)